

SYSC 5108 Exam Study: Assignment 4, Question 2
Markov chain sender example: marginals, revisits, and time averages
Source question: SYSC 5108 W26 Assignment 4.pdf, Question 2

Question Summary

Question 2 refers to the example on Slide 10 of Chapter 6. The sender has a two-state Markov chain with transition matrix

$$P = \begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix}.$$

Interpretation:

- State 0: the sender is still trying to send the message.
- State 1: the message has been sent successfully.

The chain starts in state 0 at time $n = 0$, so the initial distribution is

$$\pi(0) = [1 \quad 0].$$

State 1 is absorbing, because once the message is sent successfully, the sender stays in state 1:

$$P_{11} = 1.$$

Step 1: Use the Marginal Distribution Update Rule

Chapter 6 slides define the marginal distribution by

$$\pi_i(n) = P\{X_n = i\}$$

and the update rule

$$\pi^T(n+1) = \pi^T(n)P.$$

The textbook also describes a Markov chain through a transition operator and explains that repeated application of the transition matrix updates the state distribution over time.

Let

$$\pi(n) = [\pi_0(n) \quad \pi_1(n)].$$

Then

$$\pi(n+1) = \pi(n) \begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix}.$$

Part (a): Find $\pi_0(n)$ and $\pi_1(n)$

From the update rule:

$$\pi_0(n+1) = p \pi_0(n),$$

because the only way to be in state 0 at time $n+1$ is to already be in state 0 at time n and remain there.

Since

$$\pi_0(0) = 1,$$

we get

$$\pi_0(1) = p, \quad \pi_0(2) = p^2, \quad \dots$$

So by induction,

$$\boxed{\pi_0(n) = p^n.}$$

Because the probabilities must sum to 1,

$$\pi_1(n) = 1 - \pi_0(n) = 1 - p^n.$$

So

$$\boxed{\pi_1(n) = 1 - p^n.}$$

Check:

$$\pi(1) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} p & 1-p \end{bmatrix},$$

which matches the formula for $n = 1$.

Part (b): Probability the Sender Will Revisit State 0

Because state 1 is absorbing, the chain can revisit state 0 only if it stays in state 0 on the first step.

Starting from state 0:

$$P(X_1 = 0 \mid X_0 = 0) = p.$$

If the chain moves to state 1, it never returns to state 0. Therefore,

$$\boxed{P(\text{sender revisits state 0}) = p.}$$

Important distinction:

- Probability of at least one revisit: p .
- Expected number of revisits (from the regenerative method on the slide): $\eta(0) = \frac{p}{1-p}$ for $0 \leq p < 1$.

The question asks for the first quantity, not the second.

Part (c): Time Average Probabilities $\bar{\pi}_0$ and $\bar{\pi}_1$

The time-average probability of state i is

$$\bar{\pi}_i = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \pi_i(n).$$

Using part (a):

$$\bar{\pi}_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} p^n.$$

For $0 \leq p < 1$, the geometric sum is bounded:

$$\sum_{n=0}^{N-1} p^n = \frac{1 - p^N}{1 - p}.$$

Therefore

$$\bar{\pi}_0 = \lim_{N \rightarrow \infty} \frac{1 - p^N}{N(1 - p)} = 0.$$

So

$$\boxed{\bar{\pi}_0 = 0.}$$

Since

$$\pi_1(n) = 1 - p^n,$$

we get

$$\bar{\pi}_1 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} (1 - p^n).$$

This is

$$\bar{\pi}_1 = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N} \sum_{n=0}^{N-1} p^n \right) = 1 - 0 = 1.$$

So

$$\boxed{\bar{\pi}_1 = 1.}$$

Interpretation: for $0 \leq p < 1$, the sender spends only a transient amount of time in state 0, and eventually stays forever in the absorbing success state 1.

Edge case: if $p = 1$, then the message is never sent and the chain stays in state 0 forever, so

$$\bar{\pi}_0 = 1, \quad \bar{\pi}_1 = 0.$$

In the usual episodic sender example, we assume $0 \leq p < 1$.

Part (d): Probability the Sender Is in State 0 Before Success

Before the message is sent successfully, the chain has not yet entered state 1. Since the only states are 0 and 1, the sender must be in state 0.

So, for any time before successful transmission,

$$P(\text{sender is in state 0 before success}) = 1.$$

Another way to state this is:

$$X_n = 0 \quad \text{for all } n < T,$$

where T is the first time the message is sent successfully.

Step 2: Why the Results Make Sense

This chain has one transient state and one absorbing state:

- State 0 is transient when $0 \leq p < 1$, because eventually the chain leaves it.
- State 1 is absorbing and therefore recurrent.

That explains all four answers:

- $\pi_0(n) = p^n$ decays geometrically.
- $\pi_1(n) = 1 - p^n$ increases to 1.
- the probability of ever revisiting state 0 is just the probability of staying there on the next step, namely p ,
- the time average concentrates entirely on state 1.

Concise Exam Answer

With

$$P = \begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix}, \quad \pi(0) = [1 \quad 0],$$

the marginals satisfy

$$\pi_0(n+1) = p \pi_0(n), \quad \pi_0(0) = 1,$$

so

$$\pi_0(n) = p^n, \quad \pi_1(n) = 1 - p^n.$$

The sender revisits state 0 only if it remains in state 0 on the first step, so

$$P(\text{revisit state 0}) = p.$$

For $0 \leq p < 1$,

$$\bar{\pi}_0 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} p^n = 0, \quad \bar{\pi}_1 = 1.$$

Hence

$$\boxed{\bar{\pi}_0 = 0, \quad \bar{\pi}_1 = 1.}$$

Before the message is sent successfully, the sender must still be in state 0, so

$$\boxed{P(\text{sender is in state 0 before success}) = 1.}$$

Cross References Used

- Assignment: SYSC 5108 W26 Assignment 4.pdf, Question 2.
- Local submitted Assignment 4 PDF checked for comparison, Question 2 response.
- Slides: Chapter 6 slide deck, slides 6–10 on Markov chains, marginal distributions, stationary transitions, and the episodic sender example.
- Slides: Chapter 6 slide deck, slide 11 on classification of states.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 17 on Markov chains, transition operators, stochastic matrices, and stationary distributions.